

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2022

Bachelor in Computer Applications

Course Title: **Web Technology**

Code No: **CACS205** Semester: **III**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. Create the following table using the HTML elements.
3. What are the various font-related attributes in CSS? Explain with suitable example.
4. How does Tier Technology function inside the architectural framework of a web application?
5. How does a web server process and handle HTTP requests and responses with example?
6. Create a server site script to use cookie variables for the login and logout process without using databases.
7. Use server-site scripting to calculate the x using the data from two text boxes.
8. What are the new form elements introduced in HTML5 and how do they differ from previous versions?

Group C

Attempt any TWO questions[2x10=20]

9. Given an XML document that contains a list of products with the following elements: productname, category, price, and quantity, write an XSLT stylesheet that:
 - a) Filters out any products that have a quantity less than 10.
 - b) Groups the remaining products by category and creates a new XML document with a new element called "category" that contains the products within that category as child elements.
 - c) Within each category element, sort the products by price in descending order.
 - d) Transforms the element "productname" into an attribute of each product element.
 - e) Add a new element called "total-price" to each product element, which contains the total price of the product (price × quantity).
10. Make a script for the server site to display the following data by reading from a database and saving any new values that the user has updated.
11. Make a well-formed XML document as per the diagram, then use an appropriate XSD document to validate it with the following restrictions:
 - a) The username must have a minimum of 8 characters.
 - b) The character limit for posts is 5 to 8.
 - c) The password must be 8 characters long and begin with Aa-Zz.
 - d) The gender must be one of male, female, or other.
 - e) The salary should range from Rs. 25,000 to Rs. 50,000.